

Bertan Bakkaloglu

professor

基本信息

姓名: Bertan Bakkaloglu

所属院校: 亚利桑那州立大学

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年龄范围: 40岁至50岁

企业经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

模拟与数模混合电路

射频集成电路

混合信号集成技术

低功耗电路与能量管理

电力电子与电机驱动

微电子器件与制造

模拟与射频IC设计

超大规模集成电路设计

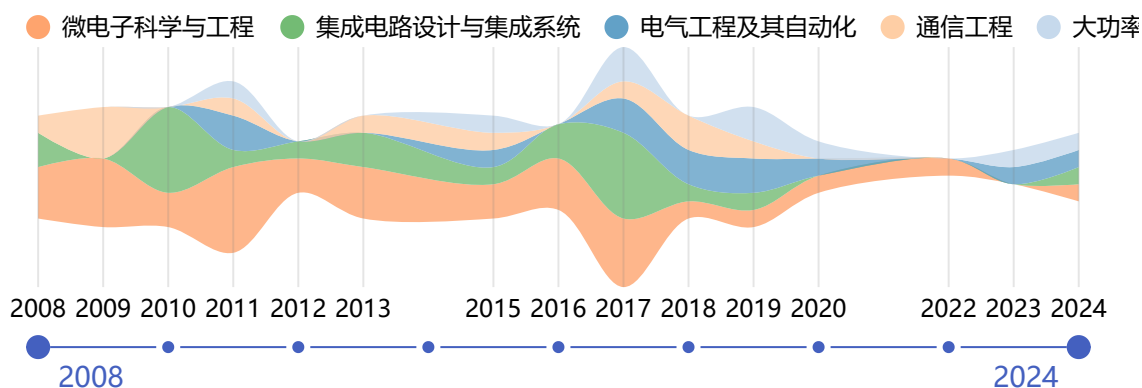
智能功率集成电路

个人简介

Bertan Bakkaloglu is the On Semiconductor Professor at the School of Electrical, Computer and Energy Engineering at ASU, which he joined in August 2004. Before that, he worked at Texas Instruments, focusing on analog, mixed - signal and RF system - on - chip development for communication transceivers. He's a technical committee member of the IEEE Radio Frequency Integrated Circuits Conference and the founding chair of the IEEE Solid State Circuits Society Phoenix Chapter. His expertise lies in RF and mixed - signal IC design, communication circuits and systems. He holds a Ph.D. in Electrical Engineering from Oregon State University (1995) and an M.S.C. from the University of

Houston (1992). Contact: , (480) 727 - 0293, ISTB4 573. Profile: <https://isearch.asu.edu/profile/749234>

近年发表的论文



5-GHz Injection-Locked Delay Cell With 10-25 ns Adjustable Group Delay in SiGe BiCMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Nazhad, Ata S.; Alizadeh, Amirreza; Frounchi, Milad; Cressler, John D.; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 0

2024

A Buck Converter Using Auxiliary-Stage With Multiple-Single-Cycle Non-Linear Control (MSCNLC) for Fast Load Transient Response

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Guo, Fan; Alevoor, Shashank; Nayak, Rakshit Dambe; Pigott, John; Goodfellow, Ryan; Bakkaloglu, Bertan

被引用次数: 0

2024

A Novel Parallel Feed-Forward Current Ripple Rejection (PFFCRR) Technique for High Load Current High PSRR nMOS LDOs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lu, Yuhong; Yen, Ting-An; Nayak, Rakshit Dambe; Alevoor, Shashank; Talele, Bhushan; Patil, Spoorti; Kunz, Keith; Bakkaloglu, Bertan

被引用次数: 0

2024

A 95.2% Efficiency DC-DC Boost Converter Using Peak Current Fast Feedback Control (PFFC) for Improved Load Transient Response

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alevoor, Shashank; Nayak, Rakshit Dambe; Talele, Bhushan; Ray, Abhishek; Rutkowski, Joseph D. D.; Stockstad, Troy; Bakkaloglu, Bertan

被引用次数: 5

2023

An Active EMI Cancellation Technique Achieving a 25-dB Reduction in Conducted EMI of LIN Drivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ray, Abhishek; Magod, Raveesh; Talele, Bhushan; Alevoor, Shashank; Mayhugh, Terry; Bakkaloglu, Bertan

A Mixed-Signal Adaptive Ripple Canceler for Switching Regulators Providing 18 dB-24 dB of Ripple Rejection up to 1 MHz

IEEE TRANSACTIONS ON POWER ELECTRONICS

Joshi, Kishan; Yang, Zhe; Fu, Chao; Mandal, Debashis; Waterfall, Gregory; Bakkaloglu, Bertan

被引用次数: 3

2020

A 5.6 μ A Wide Bandwidth, High Power Supply Rejection Linear Low-Dropout Regulator With 68 dB of PSR Up To 2 MHz

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Joshi, Kishan; Manandhar, Sanjeev; Bakkaloglu, Bertan

被引用次数: 37

2020

A 50-V Isolation, 100-MHz, 50-mW Single-Chip Junction Isolated DC-DC Converter With Self-Tuned Maximum Power Transfer Frequency

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Liu, Chengxi; Mandal, Debashis; Yao, Zhao; Sun, Ming; Todsén, Jim; Johnson, Brian; Kiaei, Sayfe; Bakkaloglu, Bertan

被引用次数: 15

2019

Low-Power/Low-Voltage Integrated CMOS Sense Resistor-Free Analog Power/Current Sensor Compatible With High-Voltage Switching DC-DC Converter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Singh, Shrikant; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 6

2019

Wideband Hybrid Envelope Tracking Modulator With Hysteretic-Controlled Three-Level Switching Converter and Slew-Rate Enhanced Linear Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mahmoudidaryan, Parisa; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 31

2019

Adaptively Biased Output Cap-Less NMOS LDO With 19 ns Settling Time

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mandal, Debashis; Desai, Chirag; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 45

2019

A 1.24 μ A Quiescent Current NMOS Low Dropout Regulator With Integrated Low-Power Oscillator-Driven Charge-Pump and Switched-Capacitor Pole Tracking Compensation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Magod, Raveesh; Bakkaloglu, Bertan; Manandhar, Sanjeev

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2018

Radiation-Tolerant Digital Multiphase Current-Mode Hysteretic Point-of-Load Regulator

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Adell, Philippe C.; Sun, Ming; Joshi, Kishan; Allen, Gregory; Yang, Zhe; Bakkaloglu, Bertan

被引用次数: 8

2018

Online Built-In Self-Test of High Switching Frequency DC-DC Converters Using Model Reference Based System Identification Techniques

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Beohar, Navankur; Malladi, Venkata N. K.; Mandal, Debashis; Ozev, Sule; Bakkaloglu, Bertan

被引用次数: 29

2018

Integrated Quasi-Circulator With RF Leakage Cancellation for Full-Duplex Wireless Transceivers

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ayati, Seyyed Amir; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

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2018

Detection Mechanisms for Unauthorized Wireless Transmissions

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Chang, Doohwang; Bhat, Ganapati; Ogras, Umit; Bakkaloglu, Bertan; Ozev, Sule

被引用次数: 5

2018

An Electrical-Stimulus-Only BIST IC for Capacitive MEMS Accelerometer Sensitivity Characterization

IEEE SENSORS JOURNAL

Ozel, Muhlis Kenan; Cheperak, Mike; Dar, Tehmoor; Kiaei, Sayfe; Bakkaloglu, Bertan; Ozev, Sule

被引用次数: 14

2017

A High Slew-Rate Adaptive Biasing Hybrid Envelope Tracking Supply Modulator for LTE Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Jing, Yue; Bakkaloglu, Bertan

被引用次数: 33

2017

A High-Voltage-Compliant Current-to-Digital Sensor for DC-DC Converters in Standard CMOS Technology

IEEE TRANSACTIONS ON POWER ELECTRONICS

Marti-Arbona, Edgar; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 6

2017

A Low-Noise Output Capacitorless Low-Dropout Regulator With a Switched-RC Bandgap Reference

IEEE TRANSACTIONS ON POWER ELECTRONICS

Magod, Raveesh; Suda, Naveen; Ivanov, Vadim; Balasingam, Ravi; Bakkaloglu, Bertan

被引用次数: 33

2017

A 50-mA 99.2% Peak Current Efficiency, 250-ns Settling Time Digital Low-Dropout Regulator With Transient Enhanced PI Controller

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lim, ChaiYong; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 28

2017

A 100-mA, 99.11% Current Efficiency, 2-mVpp Ripple Digitally Controlled LDO With Active Ripple Suppression

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Cheah, Michael; Mandal, Debashis; Bakkaloglu, Bertan; Kiaei, Sayfe

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2017

A 6 A, 93% Peak Efficiency, 4-Phase Digitally Synchronized Hysteretic Buck Converter With $\pm 1.5\%$ Frequency and $\pm 3.6\%$ Current-Sharing Error

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sun, Ming; Yang, Zhe; Joshi, Kishan; Mandal, Debashis; Adell, Philipe; Bakkaloglu, Bertan

被引用次数: 35

2017

Monitor-Based In-Field Wearout Mitigation for CMOS LC Oscillators

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Chang, Doohwang; Kitchen, Jennifer N.; Bakkaloglu, Bertan; Kiaei, Sayfe; Ozev, Sule

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2016

Design-Time Reliability Enhancement Using Hotspot Identification for RF Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chang, Doohwang; Kitchen, Jennifer N.; Bakkaloglu, Bertan; Kiaei, Sayfe; Ozev, Sule

被引用次数: 8

2016

A 65 nm Programmable ANalog Device Array (PANDA) for Analog Circuit Emulation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Suda, Naveen; Suh, Jounghyuk; Hakim, Nagib; Cao, Yu; Bakkaloglu, Bertan

被引用次数: 13

2016

Microwave Week 2015 Starts with RFIC

IEEE MICROWAVE MAGAZINE

Bakkaloglu, Bertan; Wang, Albert; Kobayashi, Kevin

被引用次数: 0

2015

Built-in Self-Calibration and Digital-Trim Technique for 14-Bit SAR ADCs Achieving ± 1 LSB INL

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Thirunakkarasu, Shankar; Bakkaloglu, Bertan

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2015

A Fast Settling Oversampled Digital Sliding-Mode DC-DC Converter

IEEE TRANSACTIONS ON POWER ELECTRONICS

Dashtestani, Ahmad; Bakkaloglu, Bertan

被引用次数: 37

2015

Developing a Broadband Amplifier for Analysis of DNA Structural and Molecular Characteristics

BIOPHYSICAL JOURNAL

Becker, Jared S.; Goryll, Michael; Bakkaloglu, Bertan; Sloan, Josh; Lambert, Josh

被引用次数: 0

2014

Tina Quach In Memoriam

IEEE MICROWAVE MAGAZINE

Bakkaloglu, Bertan; Staudinger, Joseph; Ngo, David; Barr, John T.; Schindler, Fred

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2014

Backgate Modulation Technique for Higher Efficiency Envelope Tracking

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ghajar, M. Reza; Wilk, Seth J.; Lepkowski, William; Bakkaloglu, Bertan; Thornton, Trevor J.

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Programmable Analog Device Array (PANDA): A Methodology for Transistor-Level Analog Emulation

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Suh, Jounghyuk; Suda, Naveen; Xu, Cheng; Hakim, Nagib; Cao, Yu; Bakkaloglu, Bertan

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2013

A 90-nm CMOS 5-GHz Ring-Oscillator PLL With Delay-Discriminator-Based Active Phase-Noise Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Min, Seungkee; Copani, Tino; Kiaei, Sayfe; Bakkaloglu, Bertan

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2013

Electronic-nose for detecting environmental pollutants: signal processing and analog front-end design

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Kim, Hyuntae; Konnanath, Bharatan; Sattigeri, Prasanna; Wang, Joseph; Mulchandani, Ashok; Myung, Nosang; Deshusses, Marc A.; Spanias, Andreas; Bakkaloglu, Bertan

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2012

A 280 mW, 0.07% THD+N class-D audio amplifier using a frequency-domain quantizer

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Lee, Junghan; Copani, Tino; Mayhugh, Terry, Jr.; Aravind, Bhaskar; Kiaei, Sayfe; Bakkaloglu, Bertan

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2012

Analog image recognition arrays design by using co-fabricated MOSFET and MESFETs on a 0.25 μm SOS process

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Kim, Sungho; Lepkowski, William; Thornton, Trevor J.; Bakkaloglu, Bertan

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2012

A CMOS Low-Power Transceiver With Reconfigurable Antenna Interface for Medical Implant Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Copani, Tino; Min, Seungkee; Shashidharan, Sridhar; Chakraborty, Sudipto; Stevens, Mark; Kiaei, Sayfe; Bakkaloglu, Bertan

被引用次数: 46

2011

Sub-100 μm scale on-chip inductors with CoZrTa for GHz applications

JOURNAL OF APPLIED PHYSICS

Xu, Wei; Wu, Hao; Gardner, Donald S.; Sinha, Saurabh; Dastagir, Tawab; Bakkaloglu, Bertan; Cao, Yu; Yu, Hongbin

被引用次数: 16

2011

A MEMS-Based Power-Scalable Hearing Aid Analog Front End

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Deligoz, Ilker; Naqvi, Syed R.; Copani, Tino; Kiaei, Sayfe; Bakkaloglu, Bertan; Je, Sang-Soo; Chae, Junseok

被引用次数: 33

2011

A low noise, high power efficiency supply regulator for near-field power delivery

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Harrington, Peter; Chakraborty, Sudipto; Bakkaloglu, Bertan

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2011

An SET-Free, All-Digital Controlled Point-of-Load Regulator for Next-Generation Power Systems: ADC-POL

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Adell, Philippe C.; Liu, Tao; Vermeire, Bert; Bakkaloglu, Bertan; Aveline, David

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2011

A 10 b 50 MS/s Opamp-Sharing Pipeline A/D With Current-Reuse OTAs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chandrashekar, Kailash; Bakkaloglu, Bertan

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2011

Performance Enhancement of On-Chip Inductors With Permalloy Magnetic Rings

IEEE ELECTRON DEVICE LETTERS

Xu, Wei; Sinha, Saurabh; Dastagir, Tawab; Wu, Hao; Bakkaloglu, Bertan; Gardner, Donald S.; Cao, Yu; Yu, Hongbin

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2011

A 700- μA 405-MHz All-Digital Fractional-N Frequency-Locked Loop for ISM Band Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Khalil, Waleed; Shashidharan, Sridhar; Copani, Tino; Chakraborty, Sudipto; Kiaei, Sayfe; Bakkaloglu, Bertan

被引用次数: 36

2011

Workload-Aware Neuromorphic Design of the Power Controller

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Sinha, Saurabh; Suh, Jounghyuk; Bakkaloglu, Bertan; Cao, Yu

被引用次数: 8	2011
A 133 MHz Radiation-Hardened Delay-Locked Loop	
IEEE TRANSACTIONS ON NUCLEAR SCIENCE	
Sengupta, Rajat; Vermeire, Bert; Clark, Lawrence T.; Bakkaloglu, Bertan	
被引用次数: 13	2010
A Mixed-Signal Matching State Search Based Adaptive Antenna Tuning IC	
IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS	
Song, Hang; Aberle, James T.; Bakkaloglu, Bertan	
被引用次数: 15	2010
Compact model and circuit simulations for asymmetric, independent gate FinFETs	
JOURNAL OF COMPUTATIONAL ELECTRONICS	
Dessai, Gajanan; Wu, Weimin; Bakkaloglu, Bertan; McAndrew, Colin C.; Gildenblat, Gennady	
被引用次数: 6	2010
A 20-MS/s to 40-MS/s Reconfigurable Pipeline ADC Implemented With Parallel OTA Scaling	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Chandrashekar, Kailash; Corsi, Marco; Fattaruso, John; Bakkaloglu, Bertan	
被引用次数: 11	2010
A 1 MHz Bandwidth, 6 GHz 0.18 μm CMOS Type-I $\Delta\Sigma$ Fractional-N Synthesizer for WiMAX Applications (vol 44, pg 3244, 2009)	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Hedayati, Hiva; Khalil, Waleed; Bakkaloglu, Bertan	
被引用次数: 1	2010
Combined Linear and Δ-Modulated Switch-Mode PA Supply Modulator for Polar Transmitters	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Kitchen, Jennifer N.; Chu, Connie; Kiaei, Sayfe; Bakkaloglu, Bertan	
被引用次数: 48	2009
Adaptive Blocker Rejection Continuous-Time $\Sigma\Delta$ ADC for Mobile WiMAX Applications	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Kim, Hyungseok; Lee, Junghan; Copani, Tino; Bazarjani, Seyfi; Kiaei, Sayfe; Bakkaloglu, Bertan	
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A 1 MHz Bandwidth, 6 GHz 0.18 μm CMOS Type-I $\Delta\Sigma$ Fractional-N Synthesizer for WiMAX Applications	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Hedayati, Hiva; Khalil, Waleed; Bakkaloglu, Bertan	
被引用次数: 57	2009
Design and analysis of a CMOS passive $\Delta\Sigma$ ADC for low power RF transceivers	

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Chen, Feng; Bakkaloglu, Bertan; Ramaswamy, Srinath

被引用次数: 11

2009

In situ tuning of a MEMS microphone using electrodeposited nanostructures

JOURNAL OF MICROMECHANICS AND MICROENGINEERING

Je, Sang-Soo; Harrison, Jere C.; Kozicki, Michael N.; Bakkaloglu, Bertan; Kiaei, Sayfe; Chae, Junseok

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2009

A Compact and Low-Cost MEMS Loudspeaker for Digital Hearing Aids

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Je, Sang-Soo; Rivas, Fernando; Diaz, Rodolfo E.; Kwon, Jiuk; Kim, Jeonghwan; Bakkaloglu, Bertan; Kiaei, Sayfe; Chae, Junseok

被引用次数: 43

2009

A 10 MHz Bandwidth, 2 mV Ripple PA Regulator for CDMA Transmitters

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chu, Wing-Yee; Bakkaloglu, Bertan; Kiaei, Sayfe

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2008

A linear Σ - Δ digital IF to RF DAC transmitter with embedded mixer

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Taleie, Shahin Mehdizad; Copani, Tino; Bakkaloglu, Bertan; Kiaei, Sayfe

被引用次数: 46

2008

Systematic design of supply regulated LC-tank voltage-controlled oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Xuejin; Bakkaloglu, Bertan

被引用次数: 17

2008

A CMOS Low Noise, Chopper Stabilized Low-Dropout Regulator With Current-Mode Feedback Error Amplifier

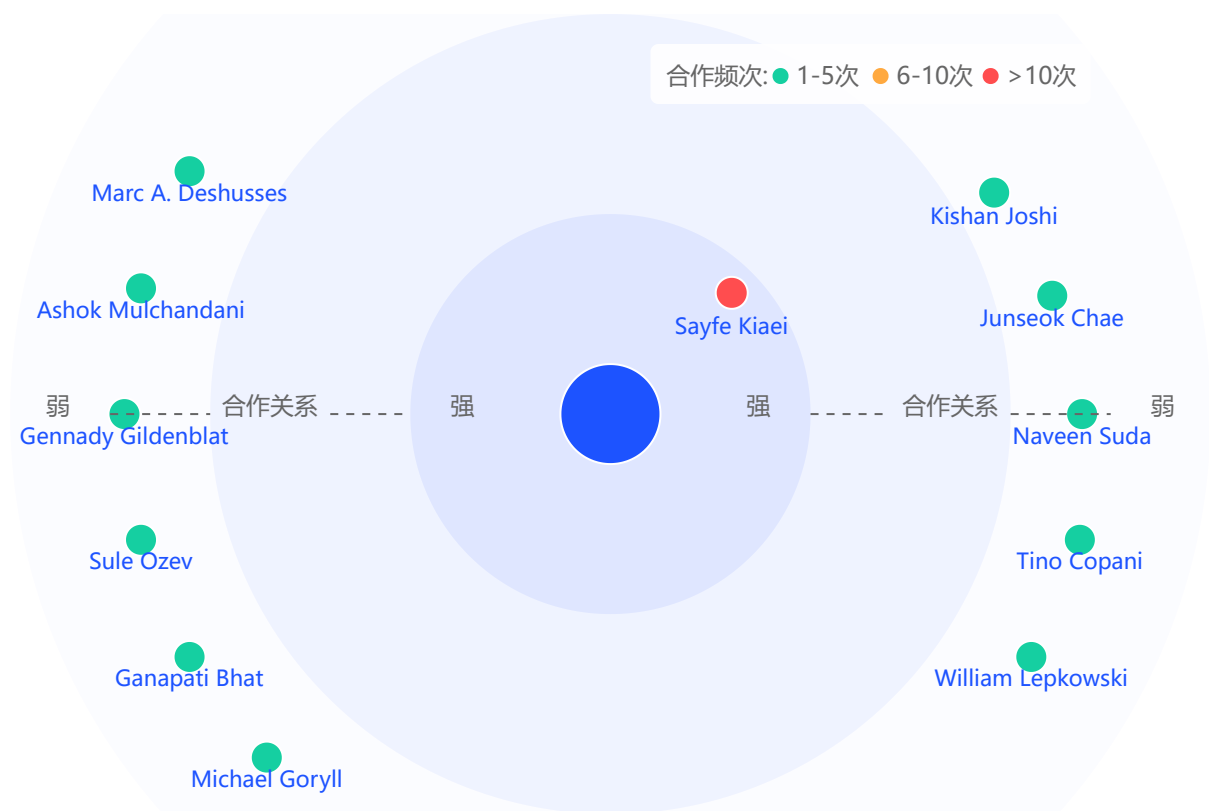
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Oh, Wonseok; Bakkaloglu, Bertan; Wang, Chris; Hoon, Siew K.

被引用次数: 43

2008

学术合作



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合作次数: 23

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亚利桑那州立大学

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还有 7 个合作者